EXPERIMENT 5

5.1 PROGRAMS:

5.1.1 //Leap Year Checker

Ans : ALGORITHM:

Step 1 : Start

Step 2 : Input year y

Step 3 : Check if y % 4 == 0

Step 4 : If yes

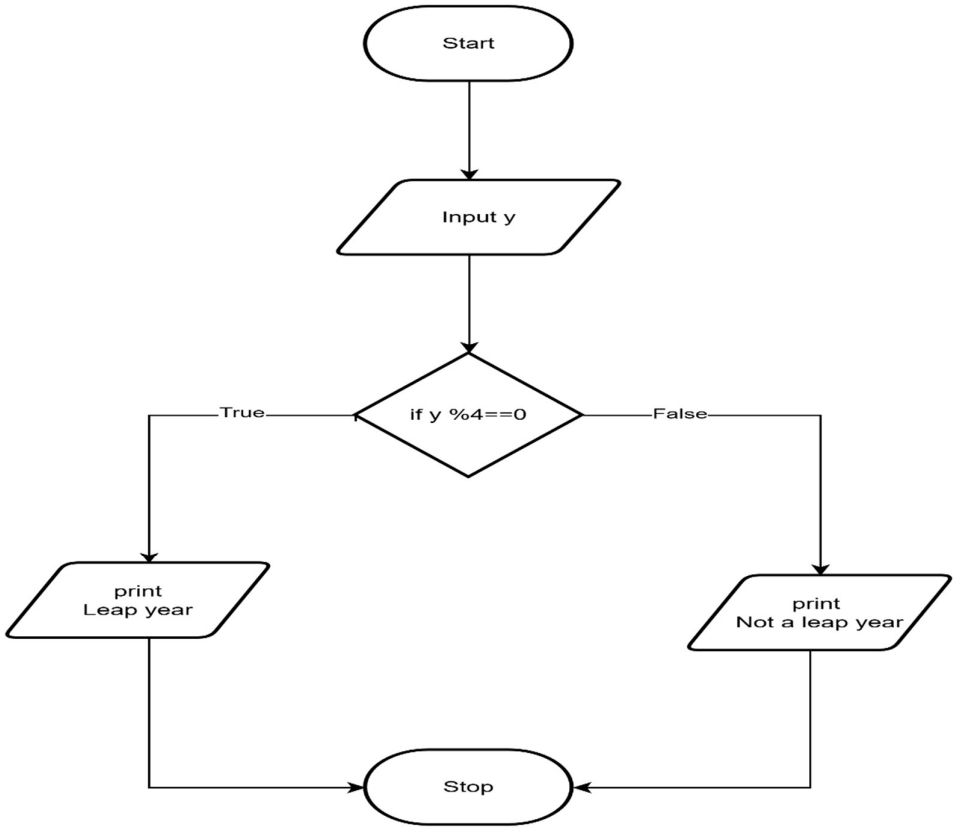
Print "Leap year"

Step 5 : Else

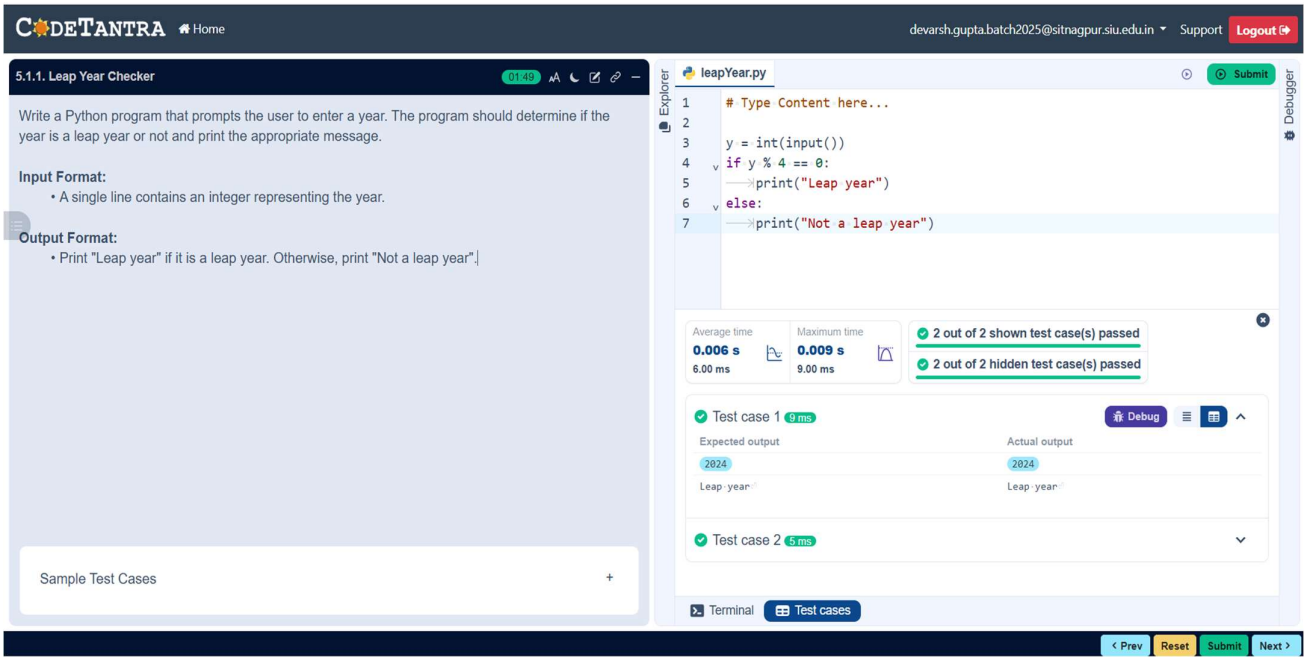
Print "Not a leap year"

Step 6 : Stop

FLOWCHART:



CODETANTRA:



5.1.2 //Student Grade Marks Based on Aggregate

Ans: ALOGORITHM

Step 1: Start

Step 2: Input four marks: a, b, c, d

Step 3: Compute total
total = a + b + c + d

Step 4: Compute percentage
percent = (total / 400) × 100

Step 5: Print total

Step 6: Print percent (up to 2 decimal places)

Step 7: Check conditions:

- **If percent > 75**

  Print Distinction

- **Else if percent ≥ 60**

  Print First Division

- **Else if percent ≥ 50**

  Print Second Division

- **Else if percent ≥ 40**

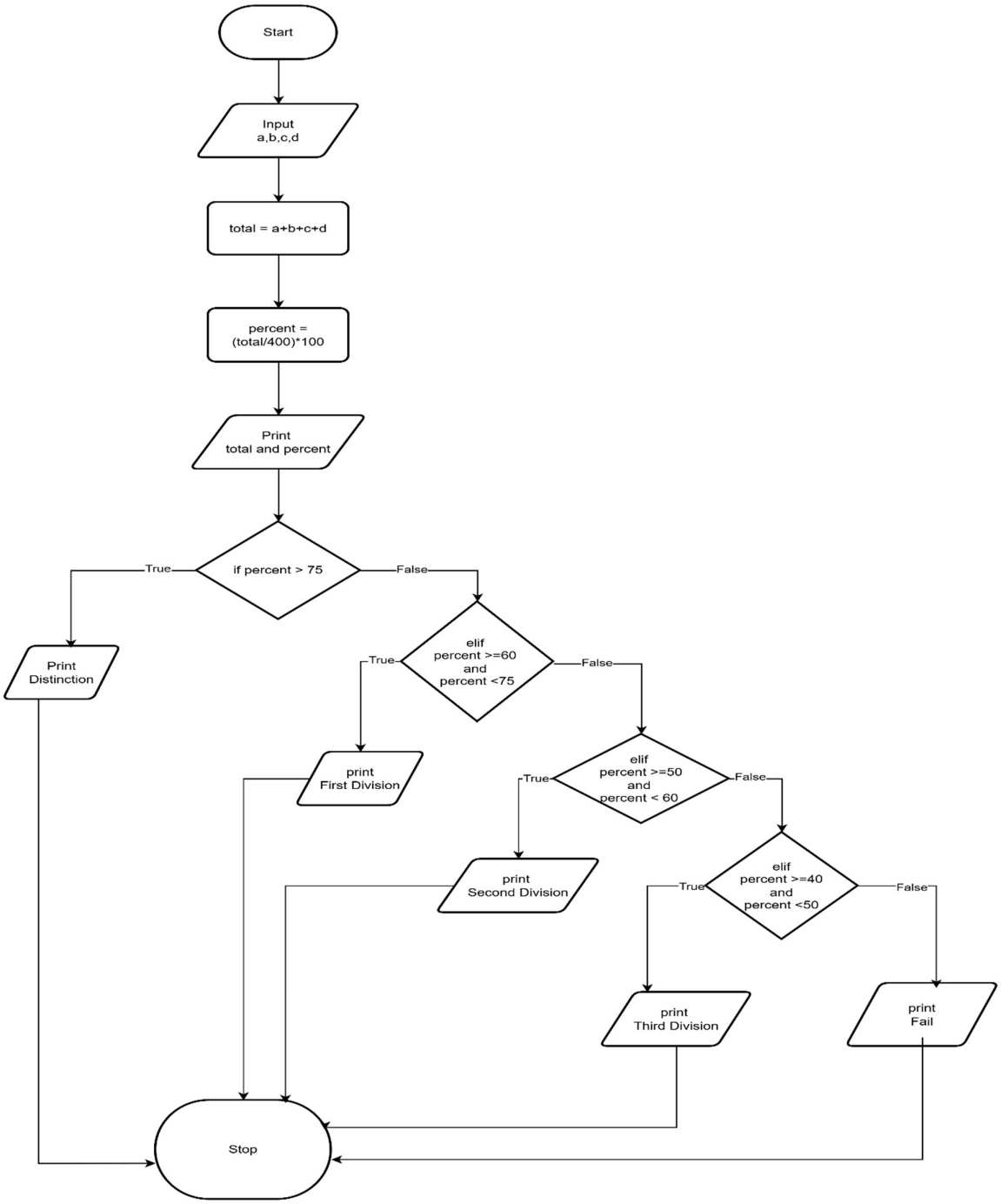
Print Third Division

- **Else**

  Print Fail

Step 8: Stop

FLOWCHART:



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5.1.2. Student Grade Based on Aggregate 13:32

Write a program to calculate the total marks, aggregate percentage, and grade of a student based on marks in four subjects. The grade is determined as follows:

- Aggregate > 75%: Distinction
- Aggregate >= 60% and < 75%: First Division
- Aggregate >= 50% and < 60%: Second Division
- Aggregate >= 40% and < 50%: Third Division
- Aggregate < 40%: Fail

Input Format:

- Four space-separated integers representing the marks in four subjects.

Output Format:

- The first line should print the total marks.
- The second line should print the aggregate percentage with two decimal places.
- The third line should print the grade.

Constraints:

- 0<=marks in each subject<=100

Sample Test Cases +

Explorer studentG... Submit Debugger

```python
# Write your code here...
a,b,c,d = map(int,input().split())
total =a+b+c+d
percent = (total/400)*100

print(total)
print(f"{percent :.2f}")

if percent > 75:
    print("Distinction")
elif percent >= 60 and percent < 75:
    print("First Division")
elif percent >= 50 and percent < 60:
    print("Second Division")
elif percent >= 40 and percent < 50:
    print("Third Division")
else:
    print("Fail")
```

Terminal Test cases

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Constraints:

- 0<=marks in each subject<=100

Sample Test Cases +

Explorer studentG... Submit Debugger

```python
# Write your code here...
a,b,c,d = map(int,input().split())
total =a+b+c+d
percent = (total/400)*100

print(total)
print(f"{percent :.2f}")

if percent > 75:
    print("Distinction")
elif percent >= 60 and percent < 75:
```

Average time 0.005 s 4.70 ms | Maximum time 0.007 s 7.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1 7 ms Debug

Expected output	Actual output
85 90 78 88	85 90 78 88
341	341
85.25	85.25
Distinction	Distinction

Test case 2 6 ms

Terminal Test cases

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